A Multi-domain inputs and task conditioning



ISIC 2018
(Dermoscopy)



BUSI-WHU
(Breast US)



Kvasir-SEG
(Endoscopy)



KUB-StoneX
(KUB X-ray)

Task identity $t \in \{0,1,2,3\}$
known domain label, GT-free

$E(t) \in R^{32}$
G FiLM γ,β

$emb_\varphi(t) \in R^{16}$
Bφ input

$onehot(t) \in R^4$
Sω input

three separate encodings → Panel B

Frozen split protocol
TRAIN 5,832 · VAL 808
TEST 1,683 (protected;
not opened for V8)

B ReMedFiLM FULL — verified inference state machine (RA-MedFiLM-BARR)

G is the backbone only

ReMedFiLM FULL = G + Rψ + Bφ + Sω · frozen modular assembly, 0 joint parameters, 0 joint backward graph

input



x
t

G · GlobalMedFiLM
Task-conditioned backbone
$E(t) \in R^{32} \rightarrow \gamma, \beta$
$\hat{F} = (1+\gamma(t))\cdot F + \beta(t)$
5 FiLM sites · 3,523,001 params



exactly one G forward

p_global
G(x,t)

GT-free ROI window
p_global ≥ 0.30 → bbox
×1.50 centre-preserving expansion
window (y0,y1,x0,x1)
no ground-truth mask used

yes ↓
action available?
no →
window is None → no

FORCED_SKIP
P_out = p_global
no ROI above 0.30
0 / 808 on VAL808

Frozen pre-refinement inputs

202-D GT-free features
Phase25B frozen cache
supplied to the wrapper
NOT a readout of G features
no ground truth at inference

task id t → embedding
Bφ's OWN 16-D embedding
202 + 16 = 218-D input
separate from G's 32-D E(t)

Bφ · Benefit Router
BARRBenefitRouterV1
$\hat{\delta} = B\varphi(f_{202}, t)$
36,801 params
trained: HuberLoss(Δ=0.05)

Resource-aware utility
$U = \hat{\delta}_final - \lambda \cdot C$
$\lambda = 1/4 \cdot C = 1051648/10270749$
$\tau_C8 = -0.0243583$
frozen decision threshold

yes (EXECUTE) ↓
U ≥ τ_C8 ?
no →

SKIP_GLOBAL
P_out = p_global
Rψ and Sω not invoked
529 / 808 on VAL808

Offline OOF producer → frozen cache

Rψ · Regional Refiner
BARRRegionalRefinerV1
4-ch [RGB, p_global crop]
1,276,401 params · raw logits
0.5·BCE + 0.5·soft Dice

p_refine (OOF cache)
256×256 refined probability
frozen out-of-fold cache
wrapper registers Rψ but
does NOT call it at runtime

Deterministic ROI fusion
inside ROI ← p_refine
outside ROI ← p_global
replacement, NO alpha blend
NO addition / learned fusion



fused probability map

P_cand
composed
candidate

Post-candidate features
16-D Phase25F frozen cache
+ 4-D one-hot task t
= 20-D Sω input
derived from P_cand offline

Sω · Post-Safety Gate
BARRPostSafetyV1
logit → a = sigmoid(logit)
1,217 params
trained: BCEWithLogitsLoss

yes
no
a ≥ 0.50 ?

FORCED_SKIP
P_out = p_global
action unavailable
0 / 808 VAL808 images
$\hat{P}$ = 1[P_out ≥ 0.50]

SKIP_GLOBAL
P_out = p_global
U < τ_C8
529 / 808 VAL808 images
$\hat{P}$ = 1[P_out ≥ 0.50]

EXECUTE_THEN_ACCEPT
P_out = P_cand
a ≥ 0.50
266 / 808 VAL808 images
$\hat{P}$ = 1[P_out ≥ 0.50]

EXECUTE_THEN_REJECT
P_out = p_global
a < 0.50
13 / 808 VAL808 images
$\hat{P}$ = 1[P_out ≥ 0.50]

C Sequential staged training (no joint end-to-end optimisation)

each stage consumes the frozen outputs of its predecessors

1 Train G
Task-conditioned GlobalMedFiLM on TRAIN 5,832.
supervised segmentation

2 Freeze G; build caches
Global predictions, ROI crops and OOF feature caches.
no joint backward graph

3 Train Rψ
4-channel ROI crop refiner, logits → p_refine.
0.5·BCE + 0.5·soft Dice

4 Paired gain labels
δ = Dice(P_cand) − Dice(p_global); OOF, cross-fitted.
δ ∈ [−1, 1]

5 Train Bφ
Regress δ from 202-D GT-free features + 16-D task embedding.
HuberLoss (Δ = 0.05)

6 Train Sω
Binary target from sign(δ) on 16-D post features + one-hot t.
BCEWith-LogitsLoss

Verification key and caveats

- G / GlobalMedFiLM — backbone and ablation baseline
- Bφ, Sω — ReMedFiLM FULL decision modules
- Rψ, ROI geometry, task conditioning
- Frozen cache supplied to the wrapper (dashed)
- Deterministic ROI replacement (no blending)

- Canonical assembly: RAMedFiLMBARR.forward_inference().
- Dashed blue / purple: p_global and ROI window reused at fusion.
- Bφ runs PRE-refinement; Sω runs POST-composition.
- The wrapper does not invoke Rψ; p_refine is an OOF cache.
- OOF boundary: Rψ / Bφ / Sω OOF=TRUE; backbone OOF=FALSE.
- No ground truth, diagnosis label or joint loss at inference.
- VAL808 state counts: 0 / 529 / 266 / 13 = 808 images.
- C1, C2, C3 controlled comparisons: NOT_ESTABLISHED.